

Create Database

Database Name:

bankdb

Collection Name:

Customers

Collection Name:

Transactions

Final structure:

bankdb

|
+ ----customers

|
+ ----transactions

The screenshot displays the MongoDB Data Explorer interface. On the left sidebar, the 'clusters' list shows 'Cluster0' expanded, with databases 'admin', 'bankdb', and 'bookflow_db' listed. Under 'bankdb', the 'customers' collection is selected. The main panel shows the 'customers' collection with 8 documents. A table lists the documents with columns: _id, ObjectId, customer_id, name, city, and account_type. The first document is Ravi Kumar, and the second is Priya Sharma. The interface also includes a query editor at the top with a 'Generate query' button and a footer with system status and copyright information.

_id	ObjectId	customer_id	name	city	account_type
1	ObjectId('6a96a3c8f36aaaf...')	101	Ravi Kumar	Hyderabad	Savings
2	ObjectId('6a96a3d5f36aaaf...')	102	Priya Sharma	Vijayawada	Current
3	ObjectId('6a96a3e0f36aaaf...')	103	Arjun R	Hyderabad	Savings
4	ObjectId('6a96a3e9f36aaaf...')	104	Sneha Rao	Warangal	Savings
5	ObjectId('6a96a3f0f36aaaf...')	105	Kiran Kumar	Guntur	Current
6	ObjectId('6a96a400f36aaaf...')	106	Anjali Singh	Hyderabad	Savings
7	ObjectId('6a96a408f36aaaf...')	107	Rahul Verma	Bengaluru	Current

PART 3: INSERT CUSTOMER DATA

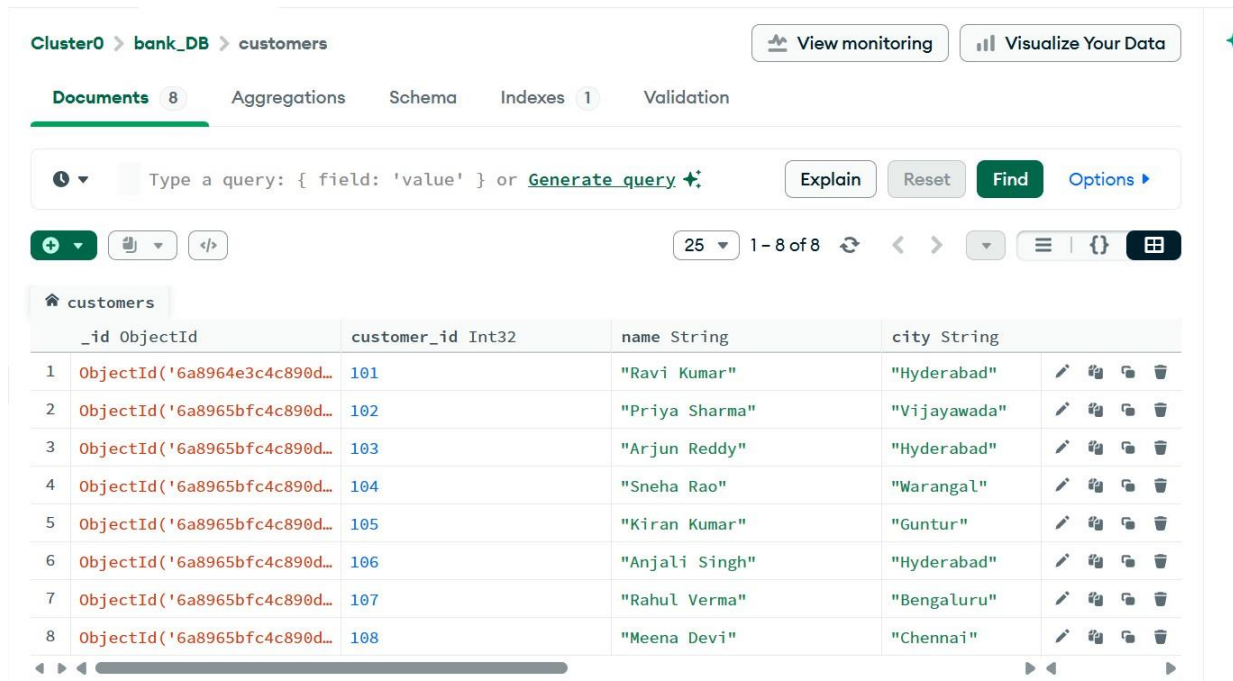
Open:

```
bankdb
|
+ ----customers
|
+ ----Documents
```

Click:

```
Add Data
|
+---- Insert Document
```

Insert the following documents one by one.



Cluster0 > bank_DB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

25 1 - 8 of 8

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	
2	ObjectId('6a8965bfc4c890d...	102	"Priya Sharma"	"Vijayawada"	
3	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	
4	ObjectId('6a8965bfc4c890d...	104	"Sneha Rao"	"Warangal"	
5	ObjectId('6a8965bfc4c890d...	105	"Kiran Kumar"	"Guntur"	
6	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	
7	ObjectId('6a8965bfc4c890d...	107	"Rahul Verma"	"Bengaluru"	
8	ObjectId('6a8965bfc4c890d...	108	"Meena Devi"	"Chennai"	

PART 4: INSERT TRANSACTION DATA

Open:

```
bankdb
|
+ ---- transactions
```

Click:

```
Add Data
|
+---- Insert Document
```

The screenshot shows the MongoDB Compass interface. At the top, the navigation bar indicates the current location: Cluster0 > bank_DB > transactions. Below this, there are tabs for Documents (15), Aggregations, Schema, Indexes (1), and Validation. A search bar is present with the placeholder text 'Type a query: { field: 'value' } or Generate query'. To the right of the search bar are buttons for 'Explain', 'Reset', 'Find', and 'Options'. Below the search bar, there are icons for adding a new document, a dropdown menu, and a code editor icon. A status bar shows '25' documents, '1 - 15 of 15' items displayed, and navigation arrows. The main area displays a table of 8 documents from the 'transactions' collection. Each document has a unique '_id' (ObjectId), a 'transaction_id' (Int32), a 'customer_id' (Int32), and a 'type' (String). The documents are numbered 1 through 8 in the first column.

	_id ObjectId	transaction_id Int32	customer_id Int32	type String
1	ObjectId('6a8966e0c4c890d...	1001	101	"Deposit"
2	ObjectId('6a8966e0c4c890d...	1002	101	"Withdrawal"
3	ObjectId('6a8966e0c4c890d...	1003	102	"Deposit"
4	ObjectId('6a8966e0c4c890d...	1004	102	"Withdrawal"
5	ObjectId('6a8966e0c4c890d...	1005	103	"Deposit"
6	ObjectId('6a8966e0c4c890d...	1006	103	"Withdrawal"
7	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"
8	ObjectId('6a8966e0c4c890d...	1008	104	"Withdrawal"

QUERY 2: FIND ACTIVE CUSTOMERS

Filter:

```
{
  "status": "Active"
}
```

The screenshot shows the MongoDB Compass interface. At the top, the breadcrumb navigation is 'Cluster0 > bank_DB > customers'. Below this, the 'Documents' tab is selected, showing 8 documents. A query filter is applied: `{ city: "Hyderabad" }`. The interface includes buttons for 'View monitoring', 'Visualize Your Data', 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. Below the query bar, there are buttons for 'ADD DATA', 'BULK', 'EXPORT CODE', and a status bar showing '25' and '1 - 3 of 3'. The main area displays a table of 3 documents, all with 'city' as 'Hyderabad' and 'status' as 'Active'.

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	   
2	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	   
3	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	   

QUERY 3: FIND CUSTOMERS WITH BALANCE GREATER THAN 100000

Filter:

```
{
  "balance": {
    "$gt": 100000
  }
}
```

The screenshot displays the MongoDB Atlas web interface. At the top, the project is 'Project 0'. The breadcrumb navigation shows 'Cluster0 > bank_DB > customers'. The 'Documents' tab is selected, showing 8 documents. A query filter is entered in the search bar: `{ "balance": { "$gt": 100000 } }`. Below the query bar, there are buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. A table of customer data is displayed below the query bar. The table has columns: '_id ObjectId', 'customer_id Int32', 'name String', and 'city String'. There are 4 rows of data. At the bottom, there are navigation controls for the table, including a page size of 25 and a total of 1 - 4 of 4 rows.

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8965bfc4c890d...	102	"Priya Sharma"	"Vijayawada"	
2	ObjectId('6a8965bfc4c890d...	104	"Sneha Rao"	"Warangal"	
3	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	
4	ObjectId('6a8965bfc4c890d...	108	"Meena Devi"	"Chennai"	

QUERY 4: FIND HYDERABAD CUSTOMERS

Filter:

```
{
  "city": "Hyderabad"
}
```

The screenshot shows the MongoDB Atlas web interface. At the top, there's a breadcrumb navigation: Cluster0 > bank_DB > customers. Below this, there are tabs for Documents (8), Aggregations, Schema, Indexes (1), and Validation. The Documents tab is active. In the center, there's a query editor with a filter: {"city": "Hyderabad"}. To the right of the query editor are buttons: Generate query, Explain, Reset, Find, and Options. Below the query editor, there's a toolbar with buttons: ADD DATA, BULK, EXPORT CODE, and a help icon. To the right of the toolbar are pagination controls: 25, 1 - 3 of 3, and navigation arrows. Below the toolbar, there's a table with the following columns: _id ObjectId, customer_id Int32, name String, and city String. The table contains three rows of data, all with city "Hyderabad".

	_id ObjectId	customer_id Int32	name String	city String
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"
2	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"
3	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"

QUERY 5: FIND SAVINGS ACCOUNT CUSTOMERS

Filter:

```
{
  "account_type": "Savings"
}
```

Cluster0 > bank_DB > customers

View monitoring
Visualize Your Data

Documents 8
Aggregations
Schema
Indexes 1
Validation

⌵
{
"account_type": "Savings"
}

Generate query ⚡
Explain
Reset
Find
Options ⌵

+ ADD DATA ⌵
BULK ⌵
EXPORT CODE
🔔

25
1 - 5 of 5
↺
↻
↷
⌵
⋮
⌵
🗑️

🏠 customers

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	✎ 🔄 📄 🗑️
2	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	✎ 🔄 📄 🗑️
3	ObjectId('6a8965bfc4c890d...	104	"Sneha Rao"	"Warangal"	✎ 🔄 📄 🗑️
4	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	✎ 🔄 📄 🗑️
5	ObjectId('6a8965bfc4c890d...	108	"Meena Devi"	"Chennai"	✎ 🔄 📄 🗑️

QUERY 6: FIND TRANSACTIONS GREATER THAN 20000

Open:

transactions

|

+ ----Documents

Filter:

```
{
  "amount": {
    "$gt": 20000
  }
}
```

Cluster0 > bank_DB > transactions [View monitoring](#) [Visualize Your Data](#)

Documents 15 Aggregations Schema Indexes 1 Validation

🕒

{
 "amount": {
 "\$gt": 20000
 }
}

[Generate query](#) ⚡ [Explain](#) [Reset](#) [Find](#) [Options](#) ▶

[+ ADD DATA](#) [BULK](#) [EXPORT CODE](#) [?](#)

25 1 - 9 of 9 [↺](#) [↻](#) [↷](#) [⌵](#) [⌶](#) [⌷](#)

🏠 transactions				
	_id ObjectId	transaction_id Int32	customer_id Int32	type String
1	ObjectId('6a8966e0c4c890d...	1001	101	"Deposit"
2	ObjectId('6a8966e0c4c890d...	1003	102	"Deposit"
3	ObjectId('6a8966e0c4c890d...	1005	103	"Deposit"
4	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"
5	ObjectId('6a8966e0c4c890d...	1008	104	"Withdrawal"

=====

QUERY 7: FIND DEPOSITS

=====

Filter:

```
{
  "type": "Deposit"
}
```

Cluster0 > bank_DB > transactions

View monitoring

Visualize Your Data

Documents 15

Aggregations

Schema

Indexes 1

Validation

🕒

{
 "type": "Deposit"
}

Generate query ⚡

Explain

Reset

Find

Options ▶

➕ ADD DATA ▼

📄 BULK ▼

🔗 EXPORT CODE ⓘ

25 ▼

1 - 8 of 8

↺

↻

↷

☰

🔗

🏠

🏠 transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	
1	ObjectId('6a8966e0c4c890d...	1001	101	"Deposit"	✎ 📄 🗑
2	ObjectId('6a8966e0c4c890d...	1003	102	"Deposit"	✎ 📄 🗑
3	ObjectId('6a8966e0c4c890d...	1005	103	"Deposit"	✎ 📄 🗑
4	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"	✎ 📄 🗑
5	ObjectId('6a8966e0c4c890d...	1009	105	"Deposit"	✎ 📄 🗑
6	ObjectId('6a8966e0c4c890d...	1010	106	"Deposit"	✎ 📄 🗑

QUERY 8: FIND ATM TRANSACTIONS

Filter:

```
{  
  "mode": "ATM"  
}
```

Cluster0

customers

transactions

+

Cluster0 > bank_DB > transactions

[View monitoring](#) [Visualize Your Data](#)

Documents 15

Aggregations

Schema

Indexes 1

Validation

🕒

{
 "mode": "ATM"
}

[Generate query](#) [↑](#)

[Explain](#)

[Reset](#)

[Find](#)

[Options](#) ▶

[ADD DATA](#) ▼

[BULK](#) ▼

[EXPORT CODE](#)

[?](#)

25 ▼

1 - 7 of 7

↺

↻

↷

⋮

{ }

🏠

🏠 transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	mode String	
1	ObjectId('6a8966e0c4c890d...	1002	101	"ATM"	✎ 📄 🗑
2	ObjectId('6a8966e0c4c890d...	1004	102	"ATM"	✎ 📄 🗑
3	ObjectId('6a8966e0c4c890d...	1006	103	"ATM"	✎ 📄 🗑
4	ObjectId('6a8966e0c4c890d...	1008	104	"ATM"	✎ 📄 🗑
5	ObjectId('6a8966e0c4c890d...	1011	106	"ATM"	✎ 📄 🗑
6	ObjectId('6a8966e0c4c890d...	1013	107	"ATM"	✎ 📄 🗑

QUERY 9: FIND DEPOSITS GREATER THAN 50000

Filter:

```
{
  "type": "Deposit",
  "amount": {
    "$gt": 50000
  }
}
```

Cluster0 > bank_DB > transactions [View monitoring](#) [Visualize Your Data](#)

[Documents](#) 15 [Aggregations](#) [Schema](#) [Indexes](#) 1 [Validation](#)

🕒

```
{
  "type": "Deposit",
  "amount": {
    "$gt": 50000
  }
}
```

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#) [?](#)

25 1 - 3 of 3 [↺](#) [↻](#) [↷](#) [⌵](#) [☰](#) [⌵](#) [⌵](#) [⌵](#)

🏠 transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	
1	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"	✎ 📄 📄 🗑️
2	ObjectId('6a8966e0c4c890d...	1010	106	"Deposit"	✎ 📄 📄 🗑️
3	ObjectId('6a8966e0c4c890d...	1014	108	"Deposit"	✎ 📄 📄 🗑️

⏪ ⏩ ⏴ ⏵

⏴ ⏵ ⏪ ⏩

QUERY 10: FIND HYDERABAD CUSTOMERS WITH HIGH BALANCE

Filter:

```
{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}
```

The screenshot displays the MongoDB Compass web interface. At the top, the breadcrumb navigation shows 'Cluster0 > bank_DB > customers'. Below this, the 'Documents' tab is active, showing 8 documents. The query editor contains the following JSON filter:

```
{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}
```

Buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options' are visible. Below the query editor, the 'customers' collection is expanded, showing a table with the following data:

	customer_id Int32	name String	city String	account_type String	
1	106	"Anjali Singh"	"Hyderabad"	"Savings"	

The interface also includes a 'View monitoring' button, a 'Visualize Your Data' button, and a 'BULK' dropdown menu. The bottom of the interface shows a pagination bar with '25' and '1-1 of 1'.

QUERY 11: FIND CUSTOMERS IN MULTIPLE CITIES

Filter:

```
{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}
```

The screenshot shows the MongoDB Compass interface. At the top, the database structure is shown: Cluster0 > bank_DB > customers. The 'Documents' tab is selected, showing 8 documents. A filter query is entered in the search bar: `{ "city": { "$in": [ "Hyderabad", "Vijayawada", "Warangal" ] } }`. Below the search bar, there are buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. The results table shows 3 documents matching the filter:

	_id ObjectId	customer_id Int32	name String	city String
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"
2	ObjectId('6a8965bfc4c890d...	102	"Priya Sharma"	"Vijayawada"
3	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"

QUERY 12: SORT CUSTOMERS BY BALANCE

Use:

Aggregations

Add Stage:

\$sort

Enter:

```
{
  "balance": -1
}
```

The screenshot shows the MongoDB Atlas Aggregations editor interface. At the top, the breadcrumb navigation indicates the path: Cluster0 > bank_DB > customers. Below this, there are tabs for Documents (8), Aggregations (selected), Schema, Indexes (1), and Validation. The main workspace is titled 'Untitled - modified' and contains a single aggregation stage named '\$sort'. The configuration for this stage shows a JSON document: { "balance": -1 }. To the right of the configuration, there are buttons for 'Generate aggregation', 'Explain', 'Run', and 'Options'. Below the configuration, there are buttons for 'SAVE', 'CREATE NEW', 'EXPORT CODE', 'PREVIEW', 'STAGES', 'TEXT', 'WIZARD', and a settings icon. The output preview section, titled 'Output preview after \$sort stage (Sample of 8 documents)', displays a sample of the sorted data. The first document shown is: { "_id": ObjectId('6a8965bfc4c890d41f9f6bce'), "customer_id": 108, "name": "Meena Devi", "city": "Chennai", "account_type": "Savings", "balance": 320000, "status": "Active" }. The second document is partially visible on the right.

QUERY 13: CALCULATE TOTAL TRANSACTION AMOUNT

Go to:

transactions

|

+ ---- Aggregations

Add Stage:

\$group

Enter:

```
{
  "_id": null,
  "total_amount": {
    "sum": "$balance"
  }
}
```

The screenshot shows the MongoDB Atlas interface for a query in the 'bank_DB' database, specifically on the 'customers' collection. The 'Aggregations' tab is selected, and a '\$group' stage is configured. The aggregation pipeline is defined as follows:

```
{
  "_id": null,
  "total_amount": {
    "sum": "$balance"
  }
}
```

The interface includes a 'Generate aggregation' button, an 'Explain' button, and a 'Run' button. Below the aggregation stage, there is a preview of the output after the '\$group' stage, showing a sample of 1 document:

```
{
  "_id": null,
  "total_amount": 1120000
}
```

The interface also shows a 'View monitoring' button and a 'Visualize Your Data' button. The bottom of the interface displays the 'STAGES' tab, indicating the current stage is '\$group'.

QUERY 14: CALCULATE AVERAGE TRANSACTION AMOUNT

Aggregation:

```
{
  "_id": null,
  "average_amount": {
    "avg": "$balance"
  }
}
```

The screenshot shows the MongoDB Compass interface for a cluster named 'Cluster0', database 'bank_DB', and collection 'customers'. The 'Aggregations' tab is selected, showing a single stage named '\$group'. The aggregation pipeline is defined as follows:

```
1 {
2   "_id": null,
3   "average_amount": {
4     "$avg": "$balance"
5   }
6 }
7 |
```

The interface includes buttons for 'Generate aggregation', 'Explain', 'Run', and 'Options'. Below the pipeline editor, there are options to 'SAVE', 'CREATE NEW', 'EXPORT CODE', and a 'PREVIEW' toggle. The 'PREVIEW' toggle is active, showing an output preview for the '\$group' stage. The preview displays a sample of 1 document with the following structure:

```
_id: null
average_amount : 140000
```


QUERY 15: TOTAL TRANSACTION AMOUNT BY TYPE

Aggregation:

```
[
  {
    "$group": {
      "_id": "$type",
      "total_amount": {
        "sum": "amount"
      }
    }
  }
]
```

The screenshot shows the MongoDB Compass interface for a database named 'bank_DB' and a collection named 'transactions'. The 'Aggregations' tab is selected, displaying the following aggregation pipeline:

```
[
  {
    "$group": {
      "_id": "$type",
      "total_amount": {
        "sum": "amount"
      }
    }
  }
]
```

Below the query editor, the 'Run' button is highlighted. The results section shows two documents:

```
{
  "_id": "Deposit",
  "total_amount": 485000
}
```

```
{
  "_id": "Withdrawal",
  "total_amount": 117000
}
```

The interface also includes navigation tabs for Documents (15), Aggregations, Schema, Indexes (1), and Validation. At the top right, there are buttons for 'View monitoring' and 'Visualize Your Data'.

QUERY 16: NUMBER OF TRANSACTIONS PER CUSTOMER

Aggregation:

```
[
  {
    "$group": {
      "_id": "$customer_id",
      "transaction_count": {
        "$sum": 1
      }
    }
  },
]
```

Untitled - modified



+ CREATE NEW



PREVIEW



Stage 1 \$group

```
1 {
2   "_id": "$customer_id",
3   "transaction_count": {
4     "$sum": 1
5   }
6 }
```

Output preview after \$group stage (Sample of 8 documents)

```
_id: 102
transaction_count : 2
```

```
{
  "$sort": {
    "transaction_count": -1
  }
}
```

ALL RESULTS



Showing 1 - 15 count results



```
_id: ObjectId('6a8966e0c4c890d41f9f6bd5')
transaction_id: 1005
customer_id: 103
type: "Deposit"
amount: 30000
mode: "UPI"
date: "2026-08-03"
status: "Success"
```

```
_id: ObjectId('6a8966e0c4c890d41f9f6bd2')
transaction_id: 1002
customer_id: 101
type: "Withdrawal"
amount: 10000
mode: "ATM"
```

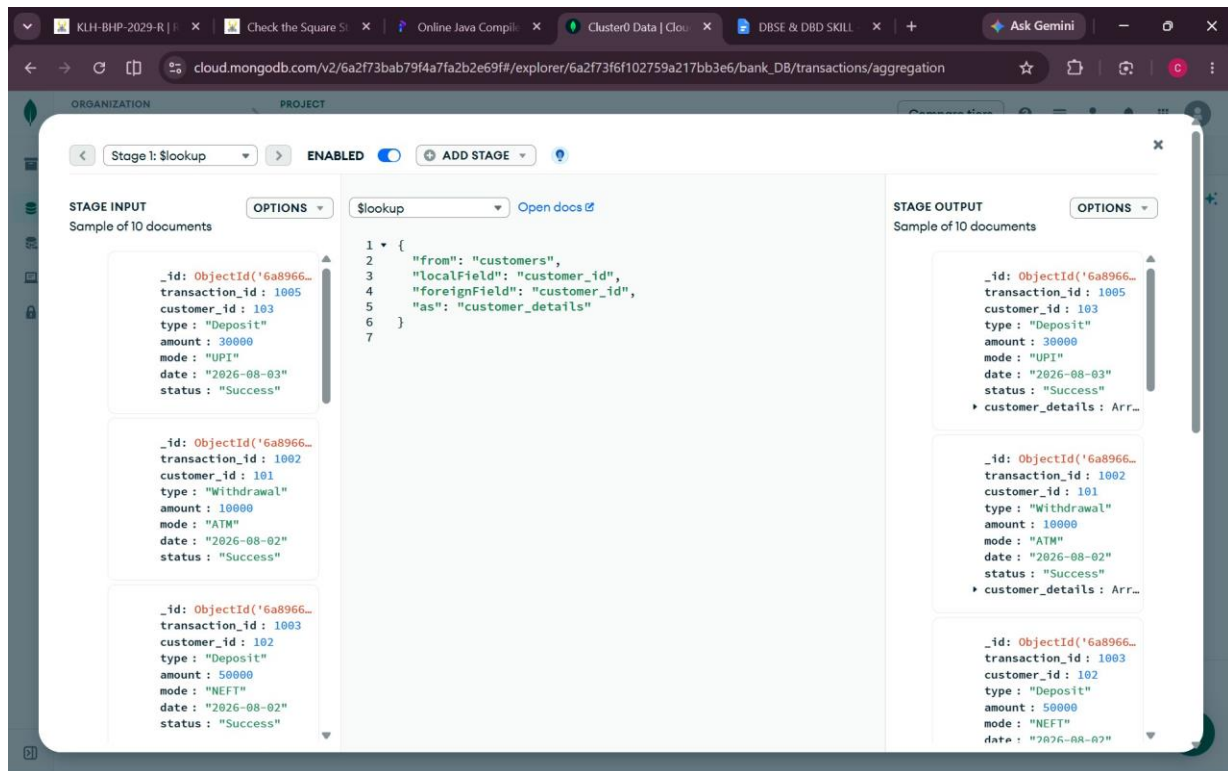
QUERY 17: CUSTOMER + TRANSACTION JOIN

Add Stage:

\$lookup

Enter:

```
{
  "from": "customers",
  "localField": "customer_id",
  "foreignField": "customer_id",
  "as": "customer_details"
}
```



Stage 1: \$lookup

ENABLED

ADD STAGE

STAGE INPUT

OPTIONS

Sample of 10 documents

_id: ObjectId('6a8966...')

transaction_id: 1005

customer_id: 103

type: "Deposit"

amount: 30000

mode: "UPI"

date: "2026-08-03"

status: "Success"

_id: ObjectId('6a8966...')

transaction_id: 1002

customer_id: 101

type: "Withdrawal"

amount: 10000

mode: "ATM"

date: "2026-08-02"

status: "Success"

_id: ObjectId('6a8966...')

transaction_id: 1003

customer_id: 102

type: "Deposit"

amount: 50000

mode: "NEFT"

date: "2026-08-02"

status: "Success"

\$lookup

Open docs

1 {

2 "from": "customers",

3 "localField": "customer_id",

4 "foreignField": "customer_id",

5 "as": "customer_details"

6 }

7 }

STAGE OUTPUT

OPTIONS

Sample of 10 documents

_id: ObjectId('6a8966...')

transaction_id: 1005

customer_id: 103

type: "Deposit"

amount: 30000

mode: "UPI"

date: "2026-08-03"

status: "Success"

customer_details: Arr...

_id: ObjectId('6a8966...')

transaction_id: 1002

customer_id: 101

type: "Withdrawal"

amount: 10000

mode: "ATM"

date: "2026-08-02"

status: "Success"

customer_details: Arr...

_id: ObjectId('6a8966...')

transaction_id: 1003

customer_id: 102

type: "Deposit"

amount: 50000

mode: "NEFT"

date: "2026-08-02"

status: "Success"

customer_details: Arr...

Go to previous stage Ctrl + Shift + <

Stage 2: \$unwind

ENABLED

ADD STAGE

STAGE INPUT

OPTIONS

Sample of 10 documents

_id: ObjectId('6a8966...')

transaction_id: 1005

customer_id: 103

type: "Deposit"

amount: 30000

mode: "UPI"

date: "2026-08-03"

status: "Success"

customer_details: Arr...

_id: ObjectId('6a8966...')

transaction_id: 1002

customer_id: 101

type: "Withdrawal"

amount: 10000

mode: "ATM"

date: "2026-08-02"

status: "Success"

customer_details: Arr...

_id: ObjectId('6a8966...')

transaction_id: 1003

customer_id: 102

type: "Deposit"

amount: 50000

mode: "NEFT"

date: "2026-08-02"

status: "Success"

customer_details: Arr...

\$unwind

Open docs

1 "\$customer_details"

STAGE OUTPUT

OPTIONS

Sample of 10 documents

_id: ObjectId('6a8966...')

transaction_id: 1005

customer_id: 103

type: "Deposit"

amount: 30000

mode: "UPI"

date: "2026-08-03"

status: "Success"

customer_details: Obj...

_id: ObjectId('6a8966...')

transaction_id: 1002

customer_id: 101

type: "Withdrawal"

amount: 10000

mode: "ATM"

date: "2026-08-02"

status: "Success"

customer_details: Obj...

_id: ObjectId('6a8966...')

transaction_id: 1003

customer_id: 102

type: "Deposit"

amount: 50000

mode: "NEFT"

date: "2026-08-02"

status: "Success"

customer_details: Obj...

QUERY 18: DISPLAY CUSTOMER NAME AND TRANSACTION AMOUNT

Use:

\$lookup

+

\$unwind

+

\$project

Pipeline:

```
[
  {
    "$lookup": {
      "from": "customers",
      "localField": "customer_id",
      "foreignField": "customer_id",
      "as": "customer"
    }
  },
  {
    "$unwind": "$customer"
  },
  {
    "$project": {
      "_id": 0,
      "transaction_id": 1,
      "customer_name": "$customer.name",
      "city": "$customer.city",
      "type": 1,
      "amount": 1,
      "mode": 1
    }
  }
]
```

Cluster0 > bank_DB > transactions

 View monitoring

 Visualize Your Data

Documents 15

Aggregations

Schema

Indexes 1

Validation



\$lookup

\$unwind

\$project



Explain

Run

Options ▶

Untitled - modified



+ CREATE NEW



 PREVIEW



▼ Stage 3 \$project



```
1 {
2   "_id": 0,
3   "transaction_id": 1,
4   "customer_name": "$customer_details.name",
5   "city": "$customer_details.city",
6   "type": 1,
7   "amount": 1,
8   "mode": 1
9 }
```

Output preview after \$project stage (Sample of 10 documents)

```
transaction_id : 1005
type : "Deposit"
amount : 30000
mode : "UPI"
customer_name : "Arjun Reddy"
city : "Hyderabad"
```

QUERY 19: TOTAL TRANSACTION AMOUNT FOR EACH CUSTOMER

Pipeline:

```
[
  {
    "$group": {
      "_id": "$customer_id",
      "total_transaction": {
        "$sum": "$amount"
      }
    }
  },
  {
    "$lookup": {
      "from": "customers",
      "localField": "_id",
      "foreignField": "customer_id",
      "as": "customer"
    }
  },
  {
    "$unwind": "$customer"
  },
  {
    "$project": {
      "_id": 0,
      "customer_id": "$_id",
      "customer_name": "$customer.name",
      "total_transaction": 1
    }
  },
  {
    "$sort": {
      "total_transaction": -1
    }
  }
]
```

The screenshot displays the MongoDB Atlas pipeline editor interface. The pipeline is titled "Stage 5: \$sort" and is currently enabled. The stages listed are:

- Stage 1: \$group
- Stage 2: \$lookup
- Stage 3: \$unwind
- Stage 4: \$project
- Stage 5: \$sort (selected)

The "\$sort" stage is configured with the sort key "total_transaction" and the sort order "-1" (descending). The stage output shows a sample of 8 documents, sorted by total transaction amount in descending order:

total_transaction	customer_id	customer_name
180	108	Meena Devi
120	104	Sneha Rao
95	106	Anjali Singh
65	102	Priya Sharma
52	107	Rahul Verma

=====

QUERY 20: COMPLEX BANK TRANSACTION REPORT

=====

PIPELINE:

```
[
  {
    "$match": {
      "status": "Active",
      "balance": {
        "$gt": 80000
      }
    }
  },
  {
    "$lookup": {
      "from": "transactions",
      "localField": "customer_id",
      "foreignField": "customer_id",
      "as": "transactions"
    }
  },
  {
    "$unwind": "$transactions"
  },
  {
    "$group": {
      "_id": "$customer_id",
      "customer_name": {
        "$first": "$name"
      },
      "city": {
        "$first": "$city"
      },
      "balance": {
        "$first": "$balance"
      },
      "total_transactions": {
        "$sum": "$transactions.amount"
      },
      "transaction_count": {
        "$sum": 1
      }
    }
  },
]
```

```

{
  "$sort": {
    "total_transactions": -1
  }
}
]

```

The screenshot displays a data pipeline configuration window. At the top, a dropdown menu shows 'Stage 5: \$sort' selected. Below it, a list of stages is visible, with 'Stage 5: \$sort' highlighted. The main configuration area shows the '\$sort' operation with a dropdown menu set to '\$sort' and a 'total_transactions' field set to '-1'. The 'STAGE OUTPUT' section on the right shows a sample of 6 documents, including customer details like name, city, balance, and transaction counts.

Stage 5: \$sort

Options:

- Stage 1: \$match
- Stage 2: \$lookup
- Stage 3: \$unwind
- Stage 4: \$group
- Stage 5: \$sort**

Configuration:

\$sort

1 * {
2 "total_transactions": -1
3 }

STAGE OUTPUT

Sample of 6 documents

Document 1:

```

{
  "_id": 102,
  "customer_name": "Priya Sharma",
  "city": "Vijayawada",
  "balance": 125000,
  "total_transactions": 6,
  "transaction_count": 2
}

```

Document 2:

```

{
  "_id": 108,
  "customer_name": "Meena Devi",
  "city": "Chennai",
  "balance": 320000,
  "total_transactions": 1,
  "transaction_count": 2
}

```

Document 3:

```

{
  "_id": 104,
  "customer_name": "Sneha Rao",
  "city": "Warangal",
  "balance": 210000,
  "total_transactions": 1,
  "transaction_count": 2
}

```

Document 4:

```

{
  "_id": 106,
  "customer_name": "Anjali Singh",
  "city": "Hyderabad",
  "balance": 175000,
  "total_transactions": 9,
  "transaction_count": 2
}

```

Document 5:

```

{
  "_id": 102
}

```